

Niobium

Nb (Nb I). Strongest classified transitions below 2 μm (NIST): 40 levels, 0 lines.

Nb⁺ ionisation limit 54513.80 cm^{−1} = 6.75885 eV (183.440 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).

